

Machine Learning-Based Standard Compact Model Binning Parameter Extraction Methodology for Integrated Circuit Design of Next-Generation Semiconductor Devices

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This article proposes a neural network-based parameter extraction methodology for the Berkeley Short-Channel IGFET Model–Common Multi-Gate (BSIM–CMG) model applied to gate-all-around field effect transistors (GAAFETs), capturing both current–voltage and capacitance–voltage characteristics to support compact model library development. Conventional BSIM parameter extraction is often complex and inefficient, requiring manual intervention and significant time to cover a wide range of device dimensions and temperatures. To address these limitations, a novel binning adaptive sampling strategy is integrated into the neural network-based extraction framework to efficiently generate training data across broad device dimension ranges. In addition, the transformer-based deep neural networks are designed to output only binnable parameters, ensuring compatibility with compact model library requirements. The trained networks are tested using 3 nm node GAAFET Technology Computer Aided Design (TCAD) data under various conditions, achieving mean absolute percentage errors below 5% for both drain current and gate capacitance. Consequently, the extracted parameters are integrated with corner model parameters through binning equations. This approach results in binning models that are readily deployable in compact model libraries while significantly reducing parameter extraction time and enabling automation across a wide range of GAAFET dimensions.

1. Introduction

Compact modeling of metal-oxide-semiconductor field effect transistors (MOSFETs) is essential for the design of integrated circuits (ICs), which are composed of numerous MOSFETs. To ensure reliable simulation performance during IC design, the compact model describing electrical characteristics of MOSFETs must achieve a high level of accuracy. Furthermore, to meet the diverse requirements of digital and analog circuit designers, the compact model should cover a wide range of structural dimensions, such as gate length and width. To address this, the industry-standard compact model formulations adopted by the Compact Model Coalition (CMC)^[1] provide various mechanisms to describe the electrical behavior of MOSFETs with different structural dimensions. Among them, the binning technique, although it may compromise the physical integrity of the equation-based MOSFET models, can significantly enhance the accuracy of the model library. In this method, binning parameters and the binning equation are introduced to enable

the use of different parameter values across varying structural dimensions. Notably, most compact model libraries use the binning technique for scalability and increasing accuracy.^[2]

Among the industry-standard compact models, the Berkeley Short-Channel IGFET Model (BSIM) is widely used for modeling MOSFETs.^[3] BSIM4, which is developed for planar MOSFETs, is applicable up to process nodes around 28 nm.^[4] Beyond this, the BSIM-Common Multi-Gate (CMG) model is developed to support advanced 3D devices, such as Fin FieldEffect Transistors (FinFETs) and gate-all-around FETs (GAAFETs), which are adopted at technology nodes below 22 and 3 nm, respectively.^[5,6] As the BSIM model evolved from BSIM4 to BSIM–CMG, its computational approach shifted from a threshold-voltage-based formulation to a surface potential-based formulation, enabling more accurate physics-based modeling. However, this improvement in physical accuracy resulted in a significant increase in the number of required model parameters, necessary to capture the complex behavior of advanced devices.^[7] Consequently, the substantial increase in binning parameters has made the construction of compact model libraries for advanced MOSFETs, including GAAFETs, increasingly time-consuming

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and labor-intensive.^[8] Furthermore, the quality of the resulting model libraries can vary considerably, depending on the expertise and judgment of the modeling engineers.

To address these challenges, various attempts have been made to extract model parameters using machine learning techniques. In particular, researchers have focused on enhancing the adaptability of the models to better capture structural variations in MOSFETs. Kao et al. proposed an artificial neural network (ANN)-based method for extracting BSIM–CMG parameters that can capture up to 10% variation in gate length (L_g) and effective oxide thickness (EOT) for 10 nm node FinFETs.^[9] Chen et al. analyzed the effects of different activation functions between hidden layers using a single neural network, which is applicable to L_g values up to 110 nm for 10 nm node FinFETs.^[10] In a subsequent study, they introduced multi-stage architecture and extended the model's applicability to $L_g = 389$ nm for 1.5 nm node GAAFETs.^[11] Lee et al. proposed a multi-task learning-based neural network capable of covering GAA channel width (W_{GAA}) up to ≈ 42.5 for 3 nm node GAAFETs.^[12]

However, these studies did not consider binning techniques over a wide range of L_g and W_{GAA} , since their model included nonbinnable BSIM–CMG parameters as outputs. Based on those studies, nonbinnable parameters may vary depending on structural inputs, which are incompatible with conventional binning-based modeling approaches. For integration into widely adopted compact model library development workflows, nonbinnable parameters must remain constant regardless of variations in L_g and W_{GAA} within a given technology node. To the best of our knowledge, there has been no prior research that applies neural network-based binning modeling techniques for extracting industry-standard compact model parameters and constructing model libraries. Unlike previous studies, the neural networks proposed in this work exclusively output binnable parameters and support modeling across a wide range of L_g and W_{GAA} . Accordingly, we propose a machine learning-based model parameter extraction methodology that can increase the efficiency of compact model library development in which binning techniques are essential.

Conducting such a neural network-based modeling study requires the generation of a large amount of training data. However, given the broad value range of BSIM–CMG parameters, exhaustive extraction across the entire space is highly inefficient. To overcome this limitation, we also propose a novel sampling strategy, binning adaptive sampling, which generates training data by first assigning nominal parameter values through an exponential weighting-based interpolation scheme. After data generation through this strategy, transformer-based deep neural networks (DNNs) are trained for the extraction of BSIM–CMG model parameters. Compared to conventional multi-layer perceptron (MLP) architecture, the proposed model uses a transformer encoder, which can effectively learn the relationships between input features, for model parameter extraction applications.

In the proposed transformer-based DNNs, the encoder is composed of multihead self-attention and feed-forward layers, while the decoder is implemented as a fully connected (FC) network that maps the encoder's latent vector to the output. This work marks the first application of transformer-based DNNs to BSIM–CMG parameter extraction. The inputs to the

model include L_g , W_{GAA} , temperatures, and current–voltage/capacitance–voltage (I–V/C–V) data. To account for temperature effects, temperature is explicitly included as part of the input features.^[13] The transformer encoder can capture dependencies among the structural and thermal inputs, and the decoder is particularly effective when the output dimensionality is fixed. The trained neural network is validated using Technology Computer Aided Design (TCAD) data representing 3 nm GAAFET devices, demonstrating the applicability of the approach to advanced technology nodes. By utilizing only binnable BSIM–CMG parameters as outputs, the model accurately predicts the I–V and C–V characteristics of the test data, achieving a mean absolute percentage error of less than 5% without weighting across sweeps or devices. After extracting the BSIM–CMG parameters at different L_g and W_{GAA} , they are connected with the binning equation to create binning models for compact model library fabrication. While this study adopts the BSIM–CMG model, other industry-standard compact models that support binning equations, such as the Pennsylvania State–Philips (PSP) model, can also be employed within the same framework.^[14]

The remainder of this article is organized as follows. Section 2 describes the conventional binning modeling techniques widely adopted in the industry, along with the proposed neural network-based modeling framework. Section 3 covers the generation of training data and the neural networks training process. In Section 4, the prediction errors between the trained neural networks and the TCAD test data are analyzed, followed by an explanation of the binning procedure used to connect the extracted local models. Finally, Section 5 concludes the article.

2. Conventional Binning Methodology and its Neural Network-Based Alternative

In most standard compact models, including BSIM, two representative methodologies are commonly employed to describe the electrical characteristics of MOSFETs spanning a wide range of gate lengths and widths. The first method is global model parameter extraction, and the second is binning model parameter extraction. Global model parameter extraction involves determining model parameters formulated as functions of gate lengths and widths within equation-based models derived from device physics. Compared to the binning-based approach, global model extraction is simpler and better preserves the physical interpretation inherent in the equation-based models. However, global model extraction often exhibits limited accuracy due to inherent limitations in the equation-based models, which may fail to adequately capture device characteristics resulting from diverse process variations. To overcome this limitation, the binning technique has been introduced. By adding binning parameters and applying the binning equation, the design space is divided into multiple geometry bins and optimized parameter sets are assigned to each bin. The BSIM model supports binning techniques by connecting parameter sets extracted at different gate lengths and widths, enabling continuous modeling across the design space. This enables circuit designers to simulate geometry-dependent device characteristics within circuit simulators by specifying the desired device dimensions. The following equation represents an example of the binning equation provided

by the BSIM-CMG model, which is used to connect multiple local models with different L_g and W_{GAA} to construct a bin. The BSIM-CMG parameter value at a specific L_g and W_{GAA} dimension, $P(L, W)$, is calculated as shown in Equation (1).

$$P(L, W) = P_i + \frac{10^{-6} \times P_{Li}}{L_g} + \frac{10^{-6} \times P_{Wi}}{W_{GAA}} + \frac{10^{-12} \times P_{Pi}}{L_g \times W_{GAA}} \quad (1)$$

here, L_g and W_{GAA} represent gate length and GAAFET channel width, respectively, and serve as dimensional variables for distance-based interpolation. The reference term (P_i), the L_g variation term (P_{Li}), the W_{GAA} variation term (P_{Wi}), and the cross-term (P_{Pi}) are the four binning parameters used as coefficients for calculating parameter value at a specific dimension. While the BSIM-CMG binning equation also includes the number of gate-all-around wires (N_{GAA}),^[5] it was fixed in this study because it is an integer parameter rather than a continuously varying dimension and thus intentionally omitted. As four binning parameters exist in equation,^[1] four local models with different L_g and W_{GAA} are required to construct a single bin. Based on the four binning parameter values, the BSIM-CMG parameter value at a specific L_g and W_{GAA} is calculated through a weighted average, where the weights are determined by the relative distances between local models and target model dimensions. An example of binning application is presented below.

Figure 1 illustrates an example that the entire L_g and W_{GAA} design space is divided into four binning regions. The target device is assigned to the bin corresponding to its L_g and W_{GAA} dimension, and the associated binning parameter set is used. Then, the parameter values are computed using the binning equation within the bin. By partitioning the entire design space into multiple bins, it enables more accurate prediction of electrical characteristics at unseen structural dimensions compared to global model.^[15]

Although the binning-based modeling offers higher accuracy compared to global models, it requires the prior preparation of multiple local models to serve for the binning equation, which entails considerable time and computational cost. To address this issue, this study proposes a method in which the required local model sets are predicted using the neural networks. For the points where additional parameters are needed for binning but corresponding I-V and C-V data are unavailable, the missing local models are estimated through interpolation with calibrated corner models. Then, the resulting models are integrated using the binning equation. The following figure illustrates the overall workflow of the proposed methodology.

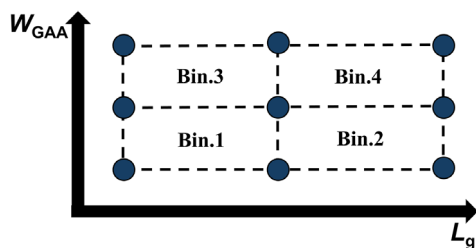


Figure 1. Example of binning models using binning equation.

As illustrated in **Figure 2**, this study consists of two main stages. In the first stage, the L_g and W_{GAA} design space is first defined, and calibration is conducted at four corner model dimensions located on its boundaries to range BSIM-CMG parameter values. Subsequently, the proposed binning adaptive sampling technique is employed to collect I-V and C-V data across variations in L_g , W_{GAA} , and temperatures.

The overall binning adaptive sampling pipeline is summarized as follows: 1) Definition of device dimension ranges; 2) nominal set determination via exponential weighting interpolation; 3) parameter variation using Monte Carlo simulation (logarithmic transform applied to fringing capacitance [CFS] and Overlap Capacitance [CGSL]); 4) each nominal set generated through physics-based filtering; and 5) merging of filtered sets into the final training datasets.

The final training data sets are then used to train the neural network, where the input is the structural and temperature conditions with I-V and C-V, and the output corresponds to the BSIM-CMG parameter values.

In the second stage, the trained neural networks are used to generate test local model sets by taking TCAD data of 3-nm GAAFET devices as input. For each bin model, four local model sets corresponding to different combinations of L_g and W_{GAA} are required. As a result, constructing multiple bins necessitates the generation of additional local models. If I-V and C-V data are available at a given point, the neural networks directly predict BSIM-CMG parameters. Otherwise, the missing local model sets are generated by interpolating corner model parameters using the binning equation. Finally, all local model sets are integrated using the binning equation to construct the complete binning model library.

3. Methodology for Training Data Preparation and Neural Network Training

3.1. Generation of Neural Network Training Data

3.1.1. User Calibration at the Four Corner L_g and W_{GAA} Dimensions

To generate the training dataset for the neural network, each BSIM-CMG parameter must be swept within a user-defined range based on the intended design space. To ensure that these ranges adequately cover the target dimensions, manual calibration is performed at the four corner L_g and W_{GAA} dimensions. However, high accuracy is not required, as the corner model parameters will be re-extracted by the neural network later. The targeted L_g and W_{GAA} coverages in this study are from 18 to 100 nm and from 23 to 60 nm, respectively. Thus, the four corner models at $(L_g, W_{GAA}) = (18, 23 \text{ nm})$, $(18, 60 \text{ nm})$, $(100, 23 \text{ nm})$, and $(100, 60 \text{ nm})$ are calibrated to establish the parameter extraction ranges. The calibration is performed based on a 3 nm GAAFET TCAD structure by extending the L_g and W_{GAA} dimensions to cover the intended design space. If hardware test patterns for compact modeling are available across various L_g and W_{GAA} values, measured electrical characteristics can be utilized in place of TCAD simulation data.

Figure 3 shows the baseline TCAD structure of the three-channel stacked GAAFET designed in this study. The geometric

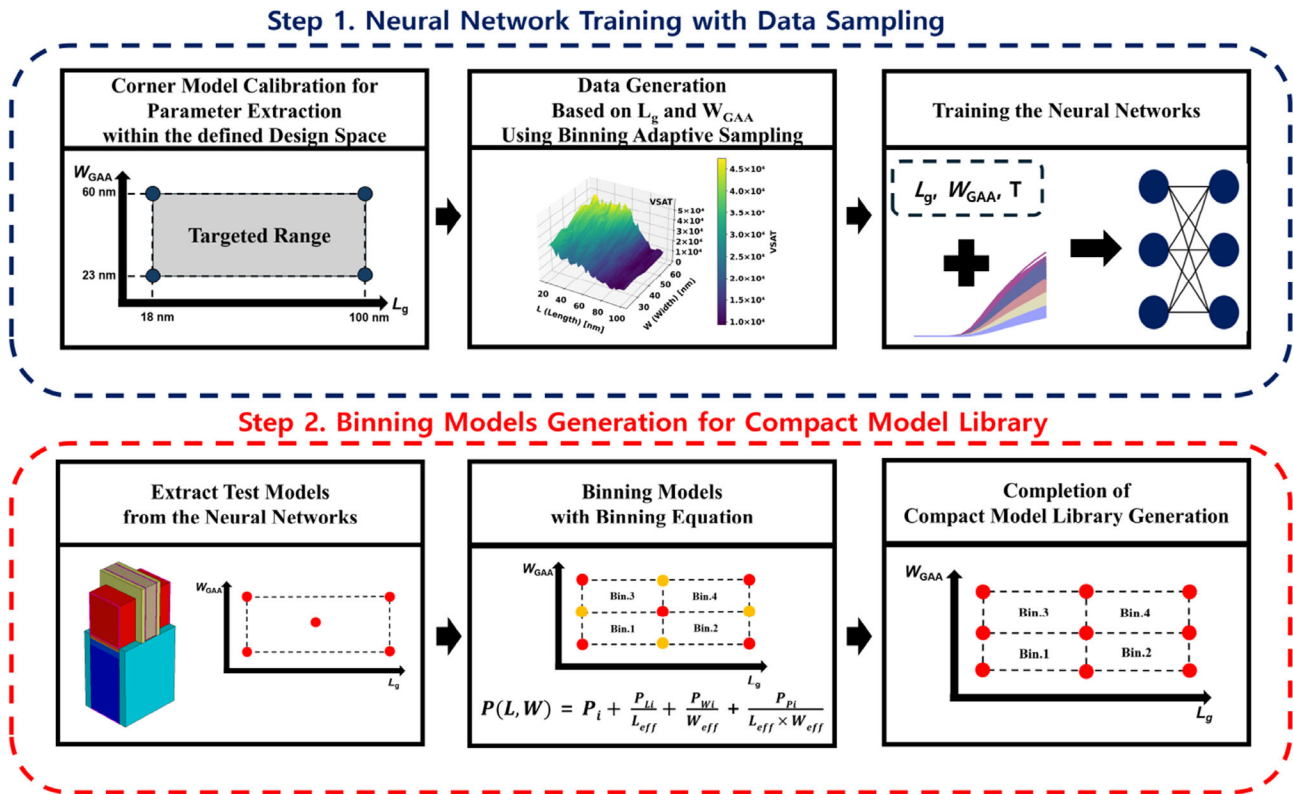


Figure 2. General workflow of the machine learning-based standard compact model binning parameter extraction and compact model library generation methodology.

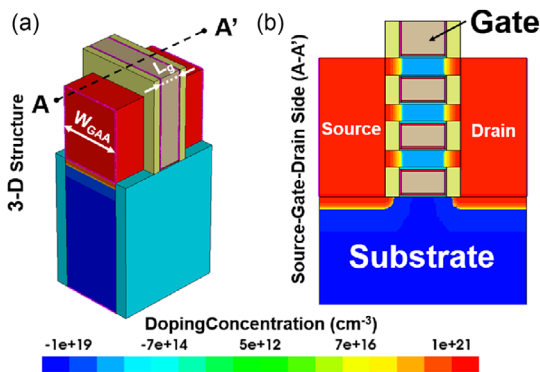


Figure 3. a) Baseline GAAFET structure with critical dimension parameter, L_g and W_{GAA} . b) Cross sectional view of GAAFET with doping scheme.

parameters, including a channel length of 18 nm and a contacted poly pitch (CPP) of 46 nm, are adopted from previous TCAD-based work that proposed a representative 3 nm node multi-channel GAAFET structure.^[16] Each silicon nanosheet channel was modeled with a width of 23 nm and a thickness of 5 nm. The Source/Drain regions are designed with a length of 18 nm, heavily doped at $5 \times 10^{20} \text{ cm}^{-3}$, while channel doping was set to a lower concentration of 10^{15} cm^{-3} . To ensure strong gate control, an equivalent oxide thickness of 0.8 nm was applied.

For electrical modeling, the Drift-Diffusion transport model was used to capture the basic carrier flow, while

Shockley–Read–Hall (SRH) and Auger recombination models were added to account for carrier recombination processes, particularly under high-injection conditions. Impact ionization was included to reflect avalanche effects under high electric fields. To simulate carrier mobility accurately, the Philips Unified Mobility model was applied to consider both phonon and Coulomb scattering. Furthermore, Thin Layer Mobility, High-Field Saturation, and Velocity Saturation models were incorporated to reflect mobility degradation and velocity-limited transport effects in the ultrathin and short-channel regime.

These models were calibrated against the measured I–V characteristics of 3 nm node GAAFET devices,^[17] ensuring the simulation reflects realistic device behavior. After achieving this calibration, TCAD data was generated across extended ranges L_g and W_{GAA} for parametric analysis.

3.1.2. Defining Nominal Values Using Exponential Weighting Interpolation

Once the modeling range of L_g and W_{GAA} is defined for BSIM–CMG parameter extraction, it is essential to perform data sampling that adequately reflects structural variations in these parameters, as the electrical characteristics of the device are highly sensitive to changes in L_g and W_{GAA} . In most of the previous works,^[9–12] BSIM–CMG parameter values are approximately determined from calibrated TCAD data at selected dimension points, and these approximate values are referred

to as nominal values that vary depending on the device dimensions. Then, the I–V and C–V data are generated by applying variations around these nominal values. In,^[9] because the targeted dimension ranges were relatively narrow, it was sufficient to use a single nominal value set without considering variations across different device dimensions. However, when the target dimension ranges become large, the corresponding BSIM–CMG parameter ranges also increase significantly. For instance, the VSAT parameter ranges from $\approx 9\,000\text{ (m}\cdot\text{s}^{-1})$ to $45\,000\text{ (m}\cdot\text{s}^{-1})$ within the targeted L_g and W_{GAA} coverage in this study. This indicates that multiple nominal value sets must be defined at specific dimension points to properly capture the parameter variation across the design space. Nevertheless, previous works^[10–12] did not specifically address how to define nominal value sets when covering a wide range of dimensions. Without multiple nominal value sets, parameter extraction must be performed over the entire parameter ranges at every dimension combination, which is highly inefficient and may lead to unphysical data that are inappropriate for neural network training. To address this issue, this study proposes binning adaptive sampling, which employs exponential weighting interpolation to reasonably define BSIM–CMG parameter nominal value sets at specific structural dimensions and subsequently generate training data by introducing variations from these nominal values. The Equation (2)–(4) show the proposed exponential weighting interpolation to determine parameter nominal values.

$$P(L, W) = P(L_{\min}, W_{\min}) \times (1 - L_{\text{scale}}) \times (1 - W_{\text{scale}}) \\ + P(L_{\min}, W_{\max}) \times (1 - L_{\text{scale}}) \times W_{\text{scale}} \\ + P(L_{\max}, W_{\min}) \times L_{\text{scale}} \times (1 - W_{\text{scale}}) \\ + P(L_{\max}, W_{\max}) \times L_{\text{scale}} \times W_{\text{scale}} \quad (2)$$

$$L_{\text{scale}} = k \times \frac{\left(e^{\alpha \times \frac{L - L_{\min}}{L_{\max} - L_{\min}}} - 1\right)}{(e^{\alpha} - 1)} \quad (3)$$

$$W_{\text{scale}} = r \times \frac{\left(e^{\beta \times \frac{W - W_{\min}}{W_{\max} - W_{\min}}} - 1\right)}{(e^{\beta} - 1)} \quad (4)$$

$L_{\min}$, $L_{\max}$, $W_{\min}$, and $W_{\max}$ in Equation (2) represent the corner dimensions. As mentioned before, in this work, the values are set to $L_{\min} = 18\text{ nm}$, $L_{\max} = 100\text{ nm}$, $W_{\min} = 23\text{ nm}$, and $W_{\max} = 60\text{ nm}$. However, the proposed methodology is flexible and can be applied to broader or narrower ranges of L_g and

W_{GAA} as needed. L_{scale} and W_{scale} in Equation (3) and (4) are defined as exponential functions, where α and β control the rate of parameter variation, while k and r adjust the overall magnitude of the variation. By the Equation (2)–(4), the parameter values at a particular dimension $P(L, W)$ are interpolated with four corner models $P(L_{\min}, W_{\min})$, $P(L_{\min}, W_{\max})$, $P(L_{\max}, W_{\min})$, $P(L_{\max}, W_{\max})$, using the exponential scaling factors L_{scale} and W_{scale} . Exponential scaling factors are introduced to account for the nonlinear scaling behavior, as the current variation with respect to dimensional changes is not uniform.

Figure 4 illustrates the distribution of the VSAT parameter extracted in this study, and the reduction in drain current as L_g increases, highlighting the importance of adopting a nonlinear form for defining nominal parameter values. As illustrated in Figure 4a, the proposed interpolation method enables the extraction of parameter values with a nonlinear scaling behavior across a wide range of L_g and W_{GAA} dimensions. Figure 4b presents the TCAD I–V characteristics ($V_{ds} = 0.05\text{ V}$) for various L_g at a fixed W_{GAA} and temperatures. It is observed that the reduction in drain current becomes less pronounced as the L_g increases. This is attributed to the reduced influence of the electric field intensity across the channel in longer devices, which consequently mitigates shortchannel effects (SCE).^[18] Thus, as demonstrated in the Equation (3) and (4), the exponential form of L_{scale} and W_{scale} provides a more suitable representation compared to the conventional linear form.

3.1.3. Data Sampling from Nominal Parameter Values

Once the nominal BSIM–CMG parameter values are determined through the exponential weighting interpolation, the BSIM–CMG parameters are varied from the nominal values through Monte Carlo Sampling with Gaussian distribution. Gaussian distribution for BSIM–CMG parameter variation was implemented in a previous related study, demonstrating its efficiency for neural network training.^[19] The variation range is set to 30% from the nominal values for most of the parameters. This range was determined based on our empirical studies, and similar empirical approaches have been adopted in previous works. For example,^[19] varied parameters by 2%–12%, while^[9] employed up to 50% based on their expertise. Although variations exceeding 30% are permitted in,^[12] such wide ranges often produced redundant or unphysical data. Consequently, the variation was limited to $\pm 30\%$, which was found to be sufficiently wide to capture realistic parameter

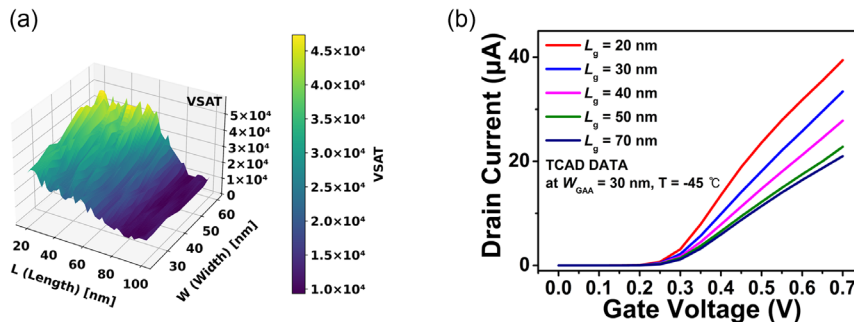


Figure 4. a) VSAT parameter distribution. b) TCAD I–V curve with various L_g .

changes while ensuring data reliability. However, some parameters are exceptions. As BSIM–CMG is the model based on surface potential, PHIG (work function) parameter has tremendous influences at both I–V and C–V characteristics. Thus, PHIG is varied with smaller ranges, following a previous related works.^[9]

Figure 5 shows the drain current data distribution at $L_g = 50$ nm, $W_{GAA} = 30$ nm, comparing results with and without the proposed binning adaptive sampling. The unsampled data in Figure 5a are extracted from a single nominal set with large variations, regardless of device dimensions, to cover wide minimum and maximum values. This approach produces a high density of outliers and unphysical curves, which can degrade training efficiency and model generalization. In contrast, the sampled data in Figure 5b are drawn from the nominal set of the targeted device dimension with smaller variations, resulting in a more balanced and physically meaningful distribution. It is worth noting that, because multiple nominal parameter sets are defined for different device dimensions, the overall coverage of the parameter space is broader while still avoiding excessive outliers. This not only prevents the generation of unphysical data but also improves learning stability by focusing on meaningful patterns.

For some capacitance parameters, CFS and CGSL, are varied after logarithmic transformation. It is because both of the parameters should be varied with large range to make reasonable gate capacitance changes, and it is ineffective to use the percentage variation from the original values without log transformation.

Figure 6 describes the distribution of C_{gg} data at $L_g = 100$ nm, $W_{GAA} = 30$ nm, and $T = -45$ °C. Figure 6a shows the C_{gg} coverage when a 30% variation is applied to CFS and CGSL without log transformation, consistent with the variation applied to most

other parameters. In contrast, Figure 6b presents the C_{gg} coverage under log-transformed CFS and CGSL with a reduced variation of 5%. Figure 5b achieves a wider C_{gg} coverage compared to Figure 5a, although only 5% variation is applied after log transformation. This can be explained in log scale, even a small variation leads to a significantly larger spread in the original value range, whereas in linear scale, the 30% variation still results in a narrower distribution around the nominal value.

$$C = \exp((1 \pm \delta) \times \ln(C_0)) \quad (5)$$

In Equation (5) and $C(0)$ is the nominal value of parameter and C is the sampled value with $\delta = 0.05$ as the variation factor. The equation describes the equation for sampling capacitance parameter after log transformation. For example, for $CFS_0 = 4.24 \times 10^{-11}$, a linear $\pm 30\%$ variation results in a limited range of 2.97×10^{-11} to 5.51×10^{-11} , leading to only minor variations in gate capacitance as mentioned in Figure 5(a). In contrast, a logarithmic-domain variation with 5% expands the range to 1.28×10^{-11} to 1.40×10^{-10} , which can sufficiently cover the expected test gate capacitance data as described in Figure 5(b). In^[12] capacitance parameters, CFS and CGSL, exhibit variations over one order of magnitude (CFS: 1×10^{-12} to 9.9×10^{-11} , CGSL: 5×10^{-11} to 1.5×10^{-10}). Such wide variation is difficult to capture adequately using simple linear-domain sampling. Therefore, adopting a logarithmic transformation is required for sampling, which naturally covers multiple orders of magnitude, ensuring adequate representation of parameter variations.

BSIM–CMG parameters are sampled via Monte Carlo simulation using a Gaussian distribution, executed through a circuit

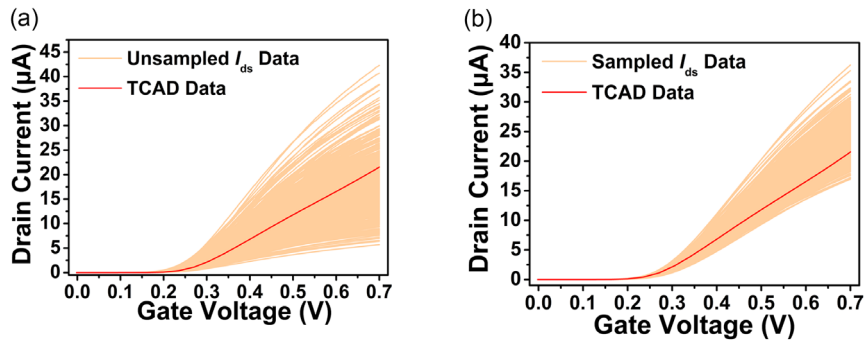


Figure 5. a) I_{ds} distribution without binning adaptive sampling. b) I_{ds} distribution after binning adaptive sampling.

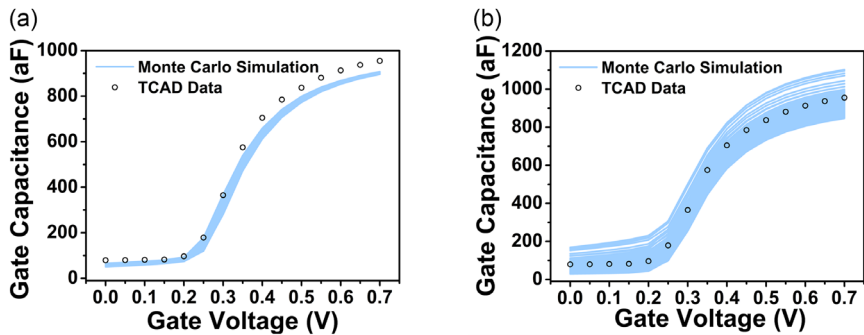


Figure 6. a) C_{gg} data without log transformation of parameters. b) C_{gg} data after log transformation of parameters.

simulator (called as SPICE) such as Synopsys' HSPICE, Cadence's SPECTRE, and so on.^[20] In contrast, extrinsic inputs such as L_g , W_{GAA} , and temperatures are sampled using latin hypercube sampling (LHS) to ensure uniform coverage of the input space.^[21] Although SPICE supports LHS sampling, applying distinct nominal values for each combination of L_g and W_{GAA} is not straightforward within a single SPICE file. Therefore, python scripts are used to implement LHS externally by generating separate SPICE files for each sampled set. This overall data generation strategy from the interpolation to sampling, referred to as binning adaptive sampling, creates $\approx 300\,000$ training data points which are used to train the neural network models.

3.2. Training of Transformer-Based Neural Networks

3.2.1. Model Structure and Output Parameters List

Compared to conventional MLP, transformer-based models can more effectively capture complex interdependencies among input parameters. Accordingly, transformer-based DNNs are employed in this study, consisting of a transformer encoder and a FC decoder. In the encoder, the input features are organized into a sequence of tokens and projected into the model dimension through an input embedding layer. The token sequence is then processed by two encoder layers, each with two attention heads (total of four). Within each encoder layer, a multihead self-attention mechanism allows each token to attend to all others in the sequence, capturing mutual dependencies among input features. The attention output is combined with the original input via a residual connection and normalized by a layer normalization operation. The normalized output is then passed through a position-wise feed-forward network that enhances representational capacity, followed by another residual connection and layer normalization. The encoder outputs a sequence of contextualized token representations, each with a dimensionality of 32, which corresponds to the latent dimension of the model. A representative 32-dimensional latent vector is selected and passed through the FC decoder to produce 23

BSIM-CMG parameter predictions. To address the inherent differences between I–V and C–V characteristics, separate neural network models are trained for each modality.

Figure 7 illustrates the overall architecture of the proposed transformer-based neural network used for both the I–V and C–V models. For the I–V model, the inputs consist of three extrinsic parameters, L_g , W_{GAA} , and temperatures, along with I–V and $\log(I)$ –V data at $V_{ds} = 0.05\text{ V}$ and V_{DD} (0.7 V). The log-transformed I–V data is included to improve learning performance in the subthreshold region, where current values are typically small.^[22] For the C–V model, the inputs include the three extrinsic parameters, C–V data at $V_{ds} = 0.05\text{ V}$, and the PHIG value obtained from the output of the I–V model. As discussed in Section 3.1.3., PHIG is a critical parameter influencing both I–V and C–V characteristics. To maintain consistency across models, the PHIG value predicted by the I–V model is used as an input to the C–V model. Unlike the workflow in,^[9] where PHIG is determined from the C–V model and then used in the I–V model, this study adopts the opposite approach. This design choice is based on our empirical findings that determining PHIG from the I–V model has a greater influence on improving I–V fitting accuracy, particularly in the subthreshold region, than determining it from the C–V model. In practice, the extracted PHIG values consistently fall within a narrow range (4.34–4.38 eV; **Table 1**). Since these values are directly transferred to the C–V model input, no additional inference-time constraints are required, and physical plausibility is inherently guaranteed by the I–V extraction process.

Table 1 summarizes the selected output BSIM–CMG parameters along with their respective minimum and maximum values used for neural network training. Parameters such as VSAT and VSAT1 exhibit wide variation ranges, highlighting the necessity of the binning adaptive sampling method introduced in Section 3.1.2. Importantly, all listed parameters are binnable, meaning they can be incorporated into the binning equation and can vary with respect to L_g and W_{GAA} within a single technology node. All other BSIM–CMG parameters, excluding the process-dependent parameters initially specified by the user,

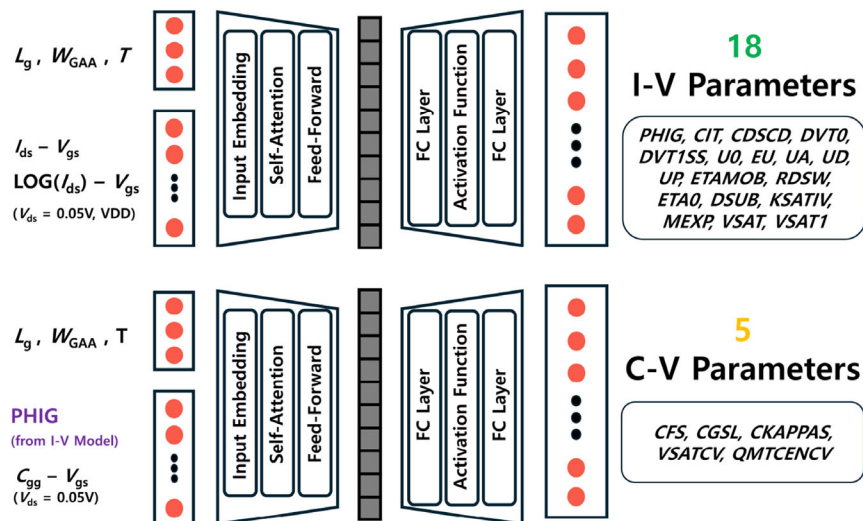


Figure 7. Structure of the proposed neural networks for parameter extraction.

Table 1. Selected output BSIM–CMG parameter lists.

Name	Min	Max
PHIG	4.34 [eV]	4.38 [eV]
CIT	2.25e-4 [$\text{F} \cdot \text{m}^{-2}$]	8.80e-4 [$\text{F} \cdot \text{m}^{-2}$]
CDSCD	2.85e-3 [$\text{F} \cdot \text{m}^{-2}$]	7.04e-3 [$\text{F} \cdot \text{m}^{-2}$]
DVT0	1.05e-2	4.54e-2
DVT1SS	0.62 [cm MV^{-1}] ^{EU}	1.42 [cm MV^{-1}] ^{EU}
U0	6.66e-3 [$\text{m}^2 \text{Vs}^{-1}$]	3.02e-2 [$\text{m}^2 \text{Vs}^{-1}$]
UP	−4.43e-6 [μm]	−1.28e-6 [μm]
EU	0.52 [cm MV^{-1}]	0.65 [cm MV^{-1}]
UA	9.14 [cm MV^{-1}] ^{EU}	21.95 [cm MV^{-1}] ^{EU}
UD	6.94 [cm MV^{-1}]	30.29 [cm MV^{-1}]
ETAMOB	29.38 [μm]	174.56 [μm]
RDSW	15.45 [$\Omega \cdot \mu\text{m}$]	70.75 [$\Omega \cdot \mu\text{m}$]
ETA0	2.14e-2	9.19e-2
DSUB	8.82e-2	2.57e-1
MEXP	1.21	2.91
KSATIV	1.11	3.82
VSAT	9140 [$\text{m} \cdot \text{s}^{-1}$]	46 262 [$\text{m} \cdot \text{s}^{-1}$]
VSAT1	2.20e5 [$\text{m} \cdot \text{s}^{-1}$]	8.01e5 [$\text{m} \cdot \text{s}^{-1}$]
CFS	1.34e-12 [$\text{F} \cdot \text{m}^{-1}$]	1.41e-10 [$\text{F} \cdot \text{m}^{-1}$]
CGSL	1.182e-11 [$\text{F} \cdot \text{m}^{-1}$]	2.713e-9 [$\text{F} \cdot \text{m}^{-1}$]
CKAPPAS	0.018 [V]	0.48 [V]
VSATCV	3805 [$\text{m} \cdot \text{s}^{-1}$]	14 280 [$\text{m} \cdot \text{s}^{-1}$]
QMTCENCV	1.55e-2	3.58e-2

are maintained at their default values throughout the training process.

3.2.2. Training Conditions

For specific training conditions, both input and output data are normalized to the range of (0, 1) using MinMaxScaler to ensure consistent feature scaling and stable model training. The dataset is split into training and validation sets with a 9:1 ratio, and the Adam optimizer with an adaptive learning rate is employed for efficient and robust optimization. A dropout layer is also included to prevent overfitting, and the loss function is defined as the

mean squared error (MSE) between the predicted and true output parameter values.

Figure 8 illustrates the training and validation loss for both the I–V and C–V models. Due to the inclusion of dropout regularization, the training process experiences slightly higher loss. In contrast, during evaluation with dropout disabled, the full network capacity is utilized, often yielding a lower validation loss. This behavior is common in models employing dropout and indicates that the model generalizes well rather than overfitting. Since the training and validation data are derived from BSIM-based SPICE simulations, whereas the final evaluation targets TCAD data, early stopping is applied when the validation loss reaches ≈ 0.003 , which is equivalent to a root MSE of around 5%. This strategy prevents overfitting to the BSIM data and ensures sufficient generalization capability, which is subsequently verified through evaluation on the TCAD dataset.

4. Results and Discussion

4.1. I–V and C–V Prediction of the Neural Networks

The trained neural networks are tested using 3 nm GAA FET TCAD data across various L_g , W_{GAA} , and temperatures conditions. Notably, since the TCAD data were not included in the training or validation datasets, this evaluation reflects the model's generalization capability. To further ensure this, the training, validation, and test sets were constructed from mutually disjoint combinations of device dimensions and temperatures. In addition, all datasets underwent the same preprocessing procedure, with the normalization parameters fitted only on the training set and subsequently applied to the validation and test sets. The neural networks results are compared with those of the nominal models, which are constructed based on the nominal values of BSIM–CMG parameters. As previously described, these nominal values are determined through exponential weighting interpolation, defined in Equation (2)–(4).

Figure 9 shows the I–V results at the condition $L_g = 20, 30, 50, 70$, and 100 nm , $W_{GAA} = 30 \text{ nm}$, $T = -45^\circ \text{C}$. The average error between NN results and TCAD data is 4.89%.

Figure 10 shows the I–V results at the condition $L_g = 20, 30, 50, 70$, and 100 nm , $W_{GAA} = 38 \text{ nm}$, $T = 5^\circ \text{C}$. The average error between NN results and TCAD data is 4.45%.

Figure 11 shows the I–V results at the condition $L_g = 20, 30, 50, 70$, and 100 nm , $W_{GAA} = 45 \text{ nm}$, $T = 65^\circ \text{C}$. The average error between NN results and TCAD data is 4.44%.

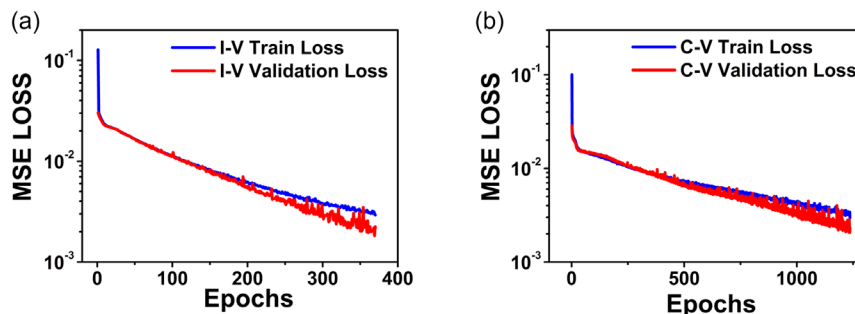


Figure 8. Train/Validation loss during the a) I–V and b) C–V model training.

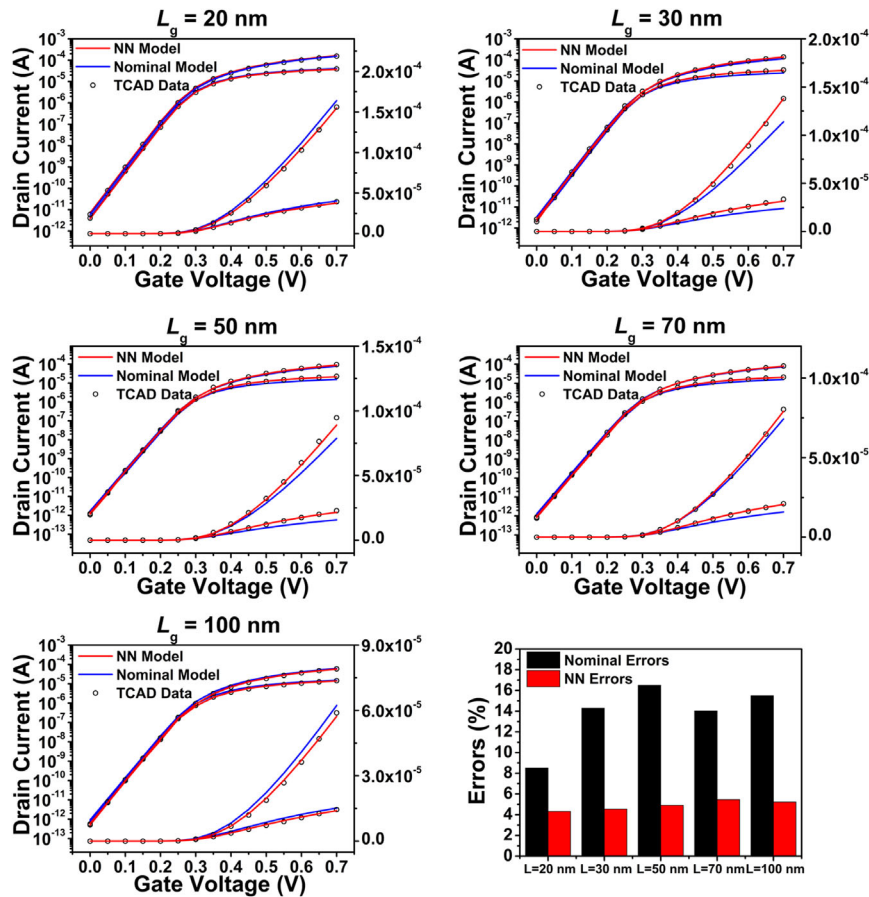


Figure 9. I–V results at $W_{GAA} = 30$ nm, $T = -45$ °C with various L_g .

Figure 12 shows the I–V results at the condition $L_g = 20, 30, 50$, and 70 , and 100 nm, $W_{GAA} = 60$ nm, $T = 80$ °C. The average error between the neural network results and TCAD data is 4.52%.

As shown in Figure 8–11, the nominal model exhibits larger errors when the target device dimensions are located further from the corner models. For example, when L_g is 50 nm and W_{GAA} is 45 nm, located in the intermediate region between the corners, the error exceeds 30%. This occurs because exponential weighting interpolation calculates nominal parameter values primarily based on the corner models, which inevitably degrades accuracy for dimensions located far from these reference points. In contrast, the proposed neural networks effectively mitigate such errors across the entire range of dimensions, thereby demonstrating their practical advantage.

Figure 13 presents the C–V errors at the device dimensions previously examined in Figure 8–11. The overall C–V error is 4.71%, with a minimum of 1.72% and a maximum of 5.89%. The slightly lower C–V accuracy compared to previous studies^[9–12] can be attributed to the intentional exclusion of non-binnable parameters, such as the physical oxide thickness (TOXP), from the model outputs. Nonbinnable parameters must remain fixed within a given technology node and should not vary with L_g , W_{GAA} , or other design variables. In contrast, previous studies incorporated these parameters as outputs to improve

fitting accuracy. This approach is incompatible with binning methodologies, where such parameters are defined as constants across all design corners. Although excluding nonbinnable parameters slightly reduces predictive accuracy, the proposed neural networks guarantee full compatibility with industry-standard compact model libraries and maintain physical interpretability—both essential aspects for practical deployment in EDA workflows.

4.2. Binning Models for Compact Model Library Generation

After confirming the accuracy of the neural networks, a series of python scripts are employed to apply the binning equation to the BSIM–CMG parameters of local models extracted at various L_g and W_{GAA} conditions. Through this process, the binning models are generated and integrated, thereby enabling the efficient construction of the final compact model library for 3 nm node GAAFETs.

Figure 14 illustrates the procedure for constructing binning models for compact model library. First, the four corner models, P(18, 23 nm), P(18, 60 nm), P(100, 23 nm), and P(100, 60 nm), are re-extracted using the neural networks, as the initially calibrated parameters obtained in Section 3.1.1 may be inaccurate. Next, a test model located between the corner models is extracted through the neural networks. In addition, extra models are

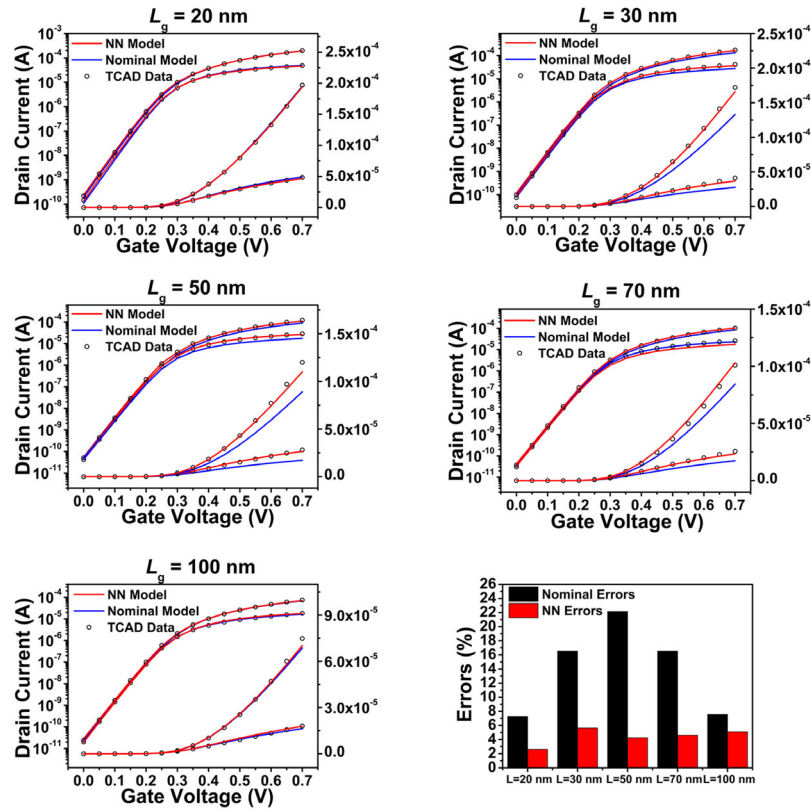


Figure 10. I-V results at $W_{GAA} = 38$ nm, $T = 5$ °C with various L_g .

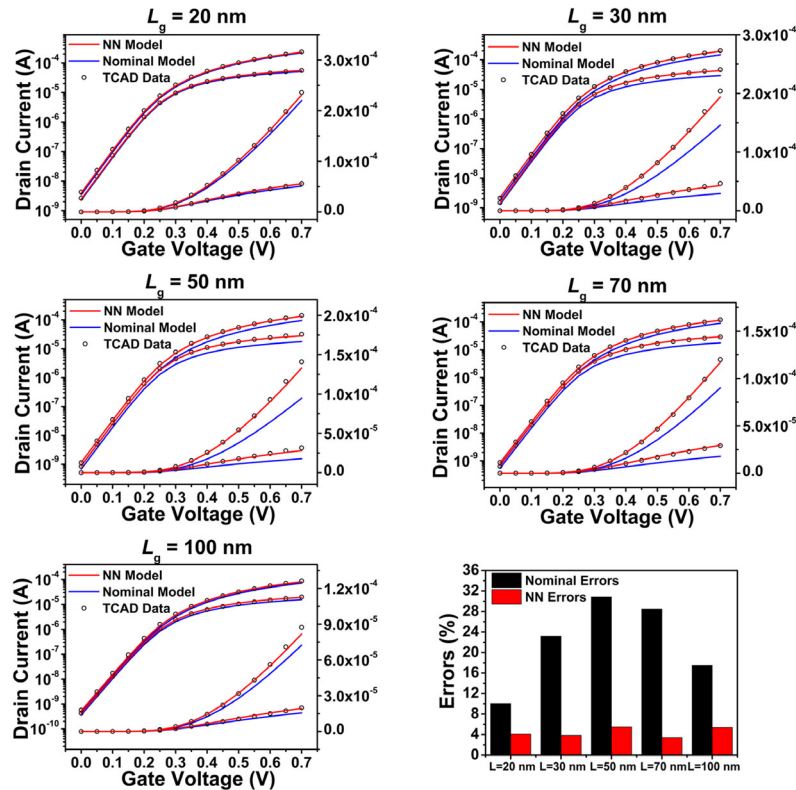


Figure 11. I-V results at $W_{GAA} = 45$ nm, $T = 65$ °C with various L_g .

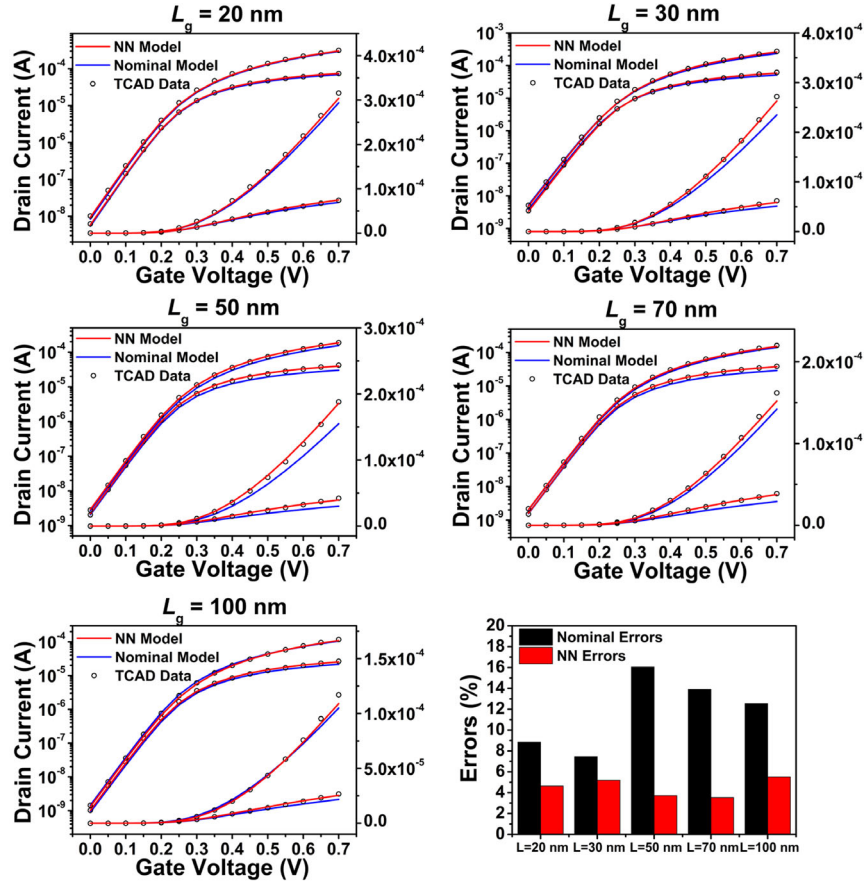


Figure 12. I–V results at $W_{GAA} = 60$ nm, $T = 80$ °C with various L_g .

extracted to ensure the required number of local models for the binning equation. Finally, by connecting all local models using the binning equations, the bins are constructed with the corresponding binning parameters. These binning models are designed to improve the prediction accuracy of electrical characteristics at black-box dimensions, where I–V or C–V data are not available.

Figure 15 shows the re-extracted corner model results obtained through the neural networks, which correspond to Step 1 in Figure 13. The temperature condition is set to 65 °C. The average errors at these corner models are 4.8% for I–V and 4.9% for C–V when using the neural networks, which represents a significant improvement compared to the initial calibration, where the errors are 9.8%. After extracting the four corner models, a test model is obtained from the neural networks, which serve as a reference point for partitioning the binning regions, as illustrated in Step 2.

To generate multiple bins, additional models are required, as described in Step 3. Since the binning equation presented in Equation (1) employs four binning parameters (P_i , P_{Li} , P_{Wi} , and P_{Pi}), four local model parameters are required per bin to solve the simultaneous equations. However, after Step 2, only five local models (four corner models and one test model) are available, which is insufficient to construct four bins, as a total of nine local models are needed. Furthermore, if I–V and C–V

data at these additional model dimensions are not available, the neural networks cannot be applied at these points. Therefore, these additional models are extracted based on the corner models, under the assumption that I–V and C–V data are not available at these dimensions, which would otherwise allow extraction through the neural networks.

Table 2 presents the U0 parameter values of the corner models extracted by the neural networks, along with the calculated U0 values of the additional models based on the corner models. By solving the binning equation using the U0 parameters of the corner models, the binning parameters are determined as follows: $P_i = 0.0326$, $P_{Li} = -3.050e-4$, $P_{Wi} = -2.800e-4$, $P_{Pi} = 2.764e-6$. P_i represents reference value of the parameter. P_{Li} accounts for the parameter variation with respect to L_g , while P_{Wi} represents the variation due to W_{GAA} . The cross-term coefficient P_{Pi} captures the interaction between both L_g and W_{GAA} . In BSIM model syntax, P_i corresponds to U0, P_{Li} to LU0, P_{Wi} to WU0, P_{Pi} to P2U0. These binning parameters connect U0 values from multiple device models which have different L_g , W_{GAA} . Assuming that the test model is at $(L_g, W_{GAA}) = (50, 45$ nm), the dimensions of four additional models where parameters should be extracted are $(L_g, W_{GAA}) = (18, 45$ nm), $(50, 23$ nm), $(50, 60$ nm), and $(100, 45$ nm). Consequently, local models at the additional dimensions can be derived based on the corner models by applying the binning equation as described in Table 2.

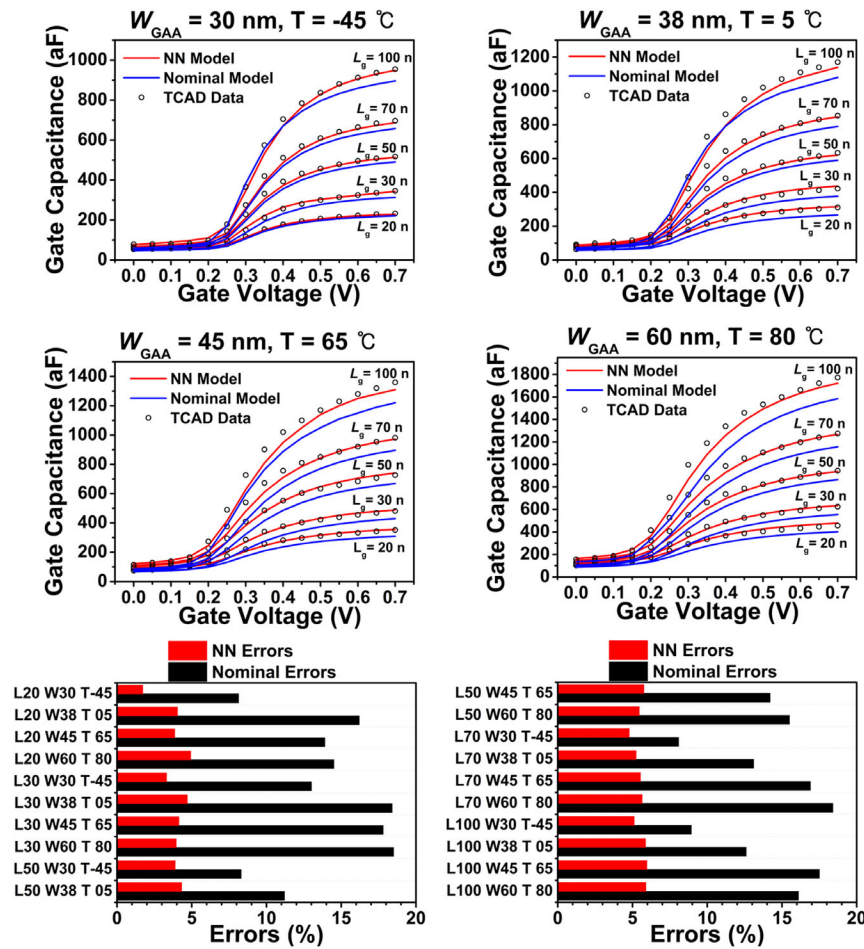


Figure 13. C-V results at various L_g , W_{GAA} and temperatures.

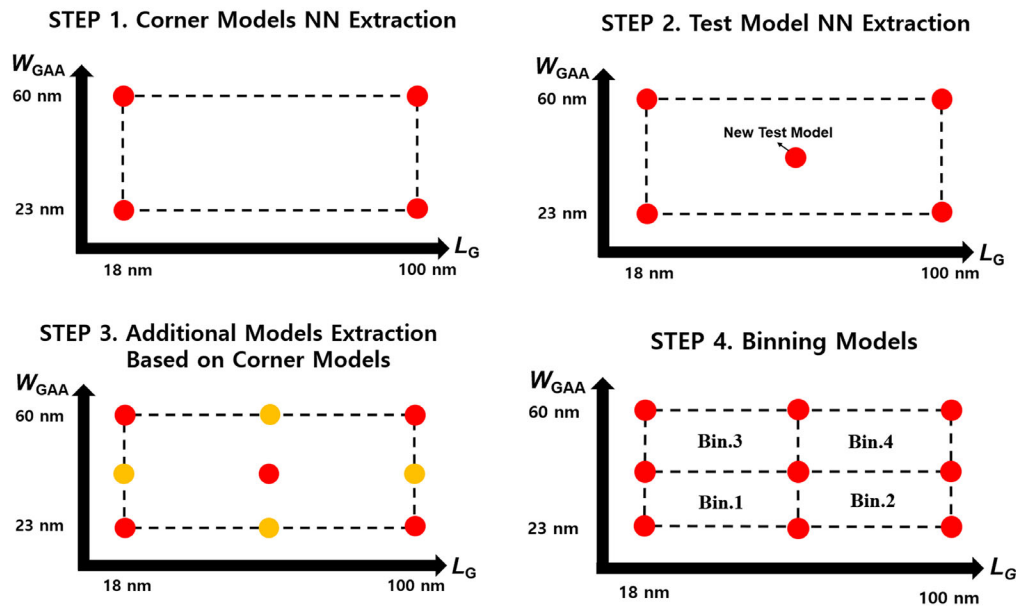


Figure 14. Steps of binning models for compact model library generation.

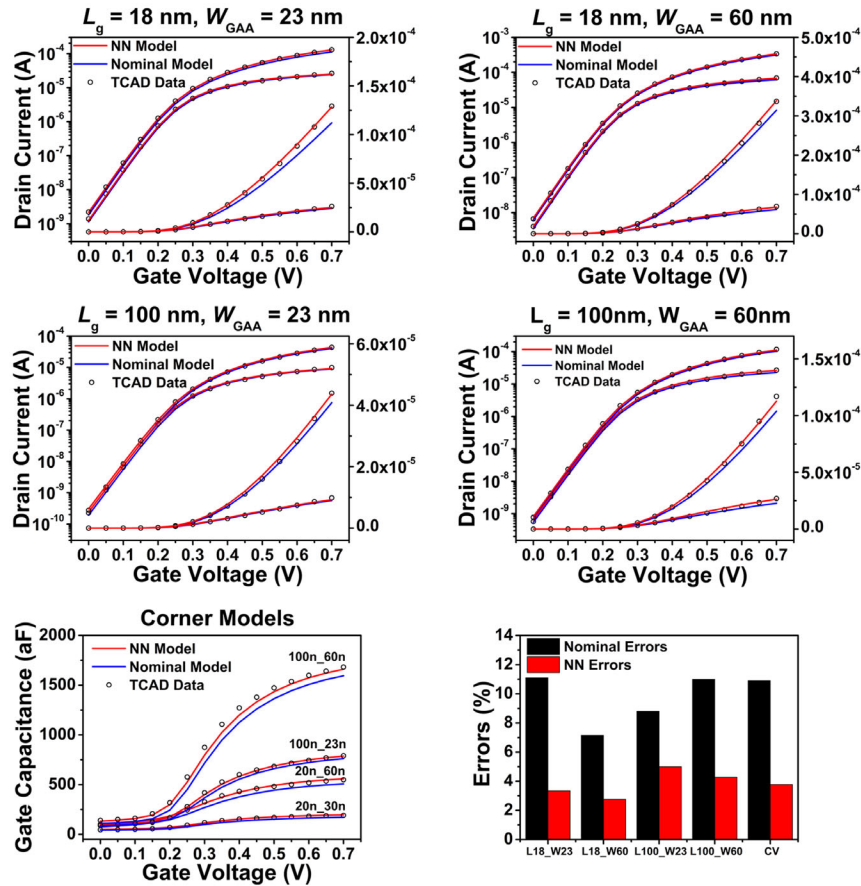


Figure 15. Re-extracted four corner models results obtained using the neural networks.

Table 2. U0 parameter values at the corner models and additional models.

Model types	Structural dimensions	U0 value
NN extracted	$L = 18 \text{ nm}, W = 23 \text{ nm}$	0.0102
Corner Models	$L = 18 \text{ nm}, W = 60 \text{ nm}$	0.0136
	$L = 100 \text{ nm}, W = 23 \text{ nm}$	0.0186
	$L = 100 \text{ nm}, W = 60 \text{ nm}$	0.0254
Calculated	$L = 18 \text{ nm}, W = 45 \text{ nm}$	0.0129
Additional Models	$L = 50 \text{ nm}, W = 23 \text{ nm}$	0.0168
	$L = 50 \text{ nm}, W = 60 \text{ nm}$	0.0228
	$L = 100 \text{ nm}, W = 45 \text{ nm}$	0.02396

After extracting all the required nine local models, final step is fabricating four bins using binning equation again. For example, Bin.3 is constructed through binning of the local models where $(L_g, W_{GAA}) = (18, 45 \text{ nm})$, $(18, 60 \text{ nm})$, $(50, 45 \text{ nm})$, and $(50, 60 \text{ nm})$.

Figure 16 shows final binning models when an extracted test model by the neural networks is at $(L_g, W_{GAA}) = (50, 45 \text{ nm})$. It can be observed that four binning models are generated from the nine local models. In order to evaluate the accuracy of the final

binning models, a local model at $(L_g, W_{GAA}) = (40, 45 \text{ nm})$ is tested, which corresponds to Bin.3. It should be noted that that binning model recognizes values in the range of $L_{\min} \leq L < L_{\max}$, and $W_{\min} \leq W < W_{\max}$. Therefore, $(L_g, W_{GAA}) = (40, 45 \text{ nm})$ is categorized into Bin.3, not Bin.1.^[5]

Figure 17 shows the errors of Bin.3 model and the unbinned model at the dimension $(L_g, W_{GAA}) = (40, 45 \text{ nm})$. The unbinned model refers to the model generated using only the corner models, without partitioning the dimensional space. The resulting errors of Bin.3 are 3.68% for I–V and 4.08% for C–V, which are significantly lower than those of the unbinned model constructed from the corner models, $(L_g, W_{GAA}) = (18, 45 \text{ nm})$, $(50, 23 \text{ nm})$, $(50, 60 \text{ nm})$, and $(100, 45 \text{ nm})$. The four local models

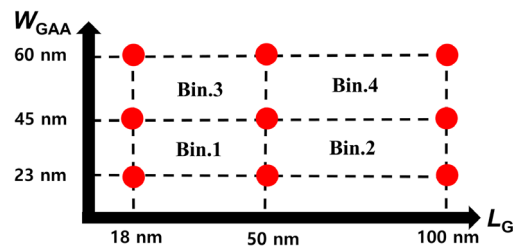


Figure 16. Final binning models assuming test model is at $(L_g, W_{GAA}) = (50, 45 \text{ nm})$.

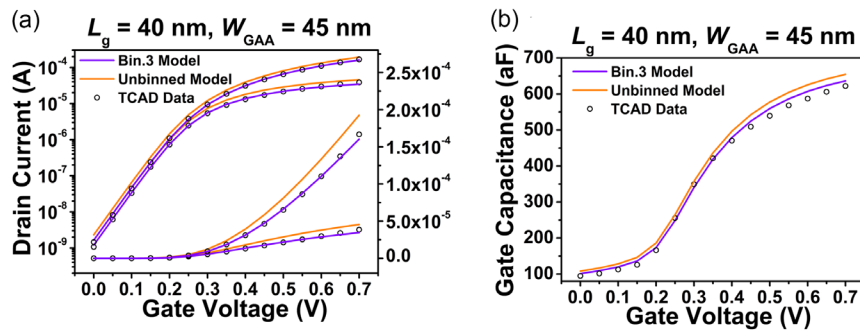


Figure 17. a) I–V and b) C–V results at a black-box dimension, $(L_g, W_{GAA}) = (40, 45 \text{ nm})$.

used to construct Bin.3 model, $(L_g, W_{GAA}) = (18, 45 \text{ nm})$, $(18, 60 \text{ nm})$, $(50, 45 \text{ nm})$ and $(50, 60 \text{ nm})$, have device dimensions that are closer to the test dimension, $(L_g, W_{GAA}) = (40, 45 \text{ nm})$, compared to the corner models used for the unbinned model. Therefore, Bin.3 model is expected to provide higher accuracy at the test dimension. These results imply that when partitioned bins are created by the proposed binning techniques, the I–V and C–V prediction accuracy can be significantly improved at black-box dimensions.

5. Conclusion

In this study, we propose a machine learning-based BSIM–CMG parameter extraction methodology and binning approach compatible with compact model library fabrication. The output BSIM–CMG parameters of the neural networks are all binnable, which represents a significant distinction from previous related studies. Furthermore, the proposed binning adaptive sampling method is utilized to set nominal BSIM–CMG parameter values across a wide range of L_g and W_{GAA} dimensions. Although N_{gaa} was fixed in this study, it could be incorporated as an additional variable in the binning scheme by extending the current bilinear interpolation in (L_g, W_{GAA}) to a trilinear form in (L_g, W_{GAA}, N_{gaa}) . Such an extension would increase the dimensionality of the sampling space and thus require substantially more data. The trained neural networks were validated using GAAFET TCAD data, resulting in $\approx 4.7\%$ errors for I–V characteristics and 4.8% errors for C–V characteristics. After extracting the required local models, the binning parameters are calculated using the binning equation provided by BSIM–CMG, and the bin models are subsequently generated. The binning models were tested at a black-box dimension, showing an error of about 4%, significantly more accurate than the model without binning. These binning models, including their binning parameters, are applicable to conventional compact model library fabrication processes because unbinnable parameters remain constant irrespective of changes in L_g and W_{GAA} . Overall, this study demonstrates the feasibility of integrating neural network-extracted local models with conventional binning methodologies, offering a practical approach to significantly enhance the accuracy and scalability of compact model libraries for advanced semiconductor devices.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

artificial neural network, Berkeley Short-Channel IGFET Model–Common Multi-Gate, compact modeling, gate-all-around field effect transistors, parameter extraction

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